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Production of human milk oligosaccharides in microbial hosts with engineered import / export

Abstract

The present invention relates to methods for the production of oligosaccharides in genetically modified bacterial host cells, as well as to the genetically modified host cells used in the methods. The genetically modified host cell comprises at least one recombinant glycosyltransferase, and at least one nucleic acid sequence coding for a protein enabling the export of the oligosaccharide.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation of U.S. application Ser. No. 15/758,653, filed on Mar. 8, 2018, which is a National Stage entry of

International Application No. PCT/EP2016/071420, filed Sep. 12, 2016, which claims priority to European Patent Application No. 15184968.4, filed Sep. 12, 2015. The disclosure of the priority applications are incorporated in their entirety herein by reference.

REFERENCE TO SEQUENCE LISTING SUBMITTED AS A COMPLIANT ASCII TEXT FILE (.txt)

(1) Pursuant to the EFS-Web legal framework and 37 CFR §§ 1.821-825 (see MPEP § 2442.03(a)), a Sequence Listing in the form of an ASCII-compliant text file (entitled “3000045-002001_Sequence_Listing_ST25.txt” created on 12 Jan. 2021, and 291,414 bytes in size) is submitted concurrently with the instant application, and the entire contents of the Sequence Listing are incorporated herein by reference.

BACKGROUND

(2) Human milk is regarded as the best diet for the development of infants. It is composed of fats, proteins, vitamins, minerals, trace elements and a complex carbohydrate mixture which comprises lactose and approximately 150 structurally diverse oligosaccharides (Human milk oligosaccharides, HMO).

DESCRIPTION OF RELATED ART

(3) Efforts to produce HMO chemically or by biotechnological approaches mainly attracted common attention due to their beneficial impact on the development of the gastrointestinal flora of infants, thus, advocating their use as nutritional additives. Besides these prebiotic properties, many other positive effects of HMO could be observed so far, expanding their field of application.

(4) However, extensive scientific studies demand pure single compounds which are hardly achievable. This is especially true for complex free neutral and acidic oligosaccharides for which competitive large-scale production processes are still lacking. (e.g. lacto-N-tetraose (Gal(β1-3)GlcNAc(β1-3)Gal(β1-4)Gluc), lacto-N-neotetraose (Gal(β1-4)GlcNAc(β1-3)Gal(β1-4)Gluc), lacto-N-fucopentaose I (Fuc(α1-2) Gal(β1-3)GlcNAc(β1-3)Gal(β1-4)Gluc) lacto-N-neofucopentaose I (Fuc(α1-2) Gal(β1-4)GlcNAc(β1-3)Gal(β1-4)Gluc) (Lacto-N-sialylpentaose a (LST-a; Neu5Ac(α2-3)Gal(β1-3)GlcNAc(β1-3)Gal(β1-4)Gluc)) The metabolic engineering of a microorganism to produce these compounds represents the most promising approach since chemical methods are rather inefficient to produce these molecules at multi-ton scale.

(5) Several fermentative approaches were already developed for the structural simpler HMOs such as 2'-fucosyllactose, 3-fucosyllactose or 3'-sialyllactose, using mainly metabolically engineered *Escherichia coli* strains.

(6) However, large-scale quantities are only achievable through boosting the oligosaccharide export out of the bacterial cell, thus, (i) enhancing the productivity and (ii) allowing the recovering of the desired oligosaccharide from the culture broth. The need for solving the export problem seems to enlarge with the size of the produced sugar. Also, with the currently available fermentation processes, upon production of more complex oligosaccharides, the problem of an unwanted export of oligosaccharide precursors from the producing cell occurs, leading to an undesirable mix of product and precursor oligosaccharides in the fermentation medium. Whereas multiple transporter proteins are known to transfer mono- or disaccharides across the membrane, hardly any knowledge exists on the transport of larger oligosaccharides (e.g., trisaccharides and larger oligosaccharides).

(7) For example, the genome of the often used fermentation model organism *E. coli* encodes more than 500 distinct transporter proteins (Busch and Saier, Crit Rev Biochem Mol Biol. 2002; 37(5):287-337). The classification of those membrane transport proteins is quite diverse and subgroups may vary in translocation mechanisms, protein structures or evolutionary origins.

(8) Classically energy-driven active transporters perform substrate movement against its concentration or electrochemical gradient, while kinetics and direction of the substrate flow through channels primarily follows such gradients. Depending on the source of energy used for the translocation, pumps can be principally divided into primary active and secondary active

transporters, exploiting metabolic energy like ATP or the electrochemical potential, respectively (Davidson and Maloney, Trends Microbiol. 2007 October; 15(10):448-55; Forrest et al, Biochim Biophys Acta. 2011 February; 1807(2):167-88). Although in-depth knowledge was achieved for several membrane proteins permitting energy generation, the import of carbohydrates and the efflux of proteins and antibacterial substances, however, keen insights into mechanistic processes or information on natural or probable substrates were gained only for a minor portion of annotated bacterial transporters so far.

(9) The *E. coli* lactose permease LacY probably represents the most intensively characterized solute transporter (Guan and Kaback, Annu Rev Biophys Biomol Struct. 2006; 35:67-91) and is a member of the large and exceptionally diverse major facilitator superfamily (MFS)—that belongs to the secondary active transporter class—transporting sugars, drugs, hydrophobic molecules, peptides, organic ions, etc. by uniport, symport or antiport (Saier et al., J Mol Microbiol Biotechnol. 1999 November; 1(2):257-79). Apart from a few exceptions a common structural feature of MFS transporters are two six-helical subdomains that transverse the cytoplasmic membrane. The existence of functionally homologous amino acid positions between related H.sup.+ -coupled MFS symporters further suggests a similar kinetic mechanism as determined for the lactose permease (Madej and Kaback, Proc Natl Acad Sci USA. 2013 Dec. 10; 110(50):E4831-8).

(10) Since decades, enormous knowledge about the import of carbohydrates into bacteria could be acquired. But regarding the export of carbohydrates, especially about molecules that are non-surface-associated, only little information is available. This is not unexpected since sugars actually depict a favourable carbon- and energy source, thus, once in the cell they shouldn't be released to a competitive environment.

(11) However, the natural function of sugar exporters probably involve the reduction of osmotic or sugar-phosphate stress which might point to a flexible substrate spectrum. Interestingly, the export of a variety of galactosides like IPTG, TMG and lactose was shown for members of the so called sugar efflux transporter family (SET), which belong to the group of MFS transporters (Liu et al., J Biol Chem. 1999 Aug. 13; 274(33):22977-84; Liu et al., Mol Microbiol. 1999 March; 31(6):1845-51).

(12) The *E. coli* transport protein SetA was even described to transfer the human milk oligosaccharide 3-fucosyllactose resulting in an improved production of said compound during fermentation of a recombinant *E. coli* strain overexpressing setA (see applicant's international patent application WO 2010/142305). Similarly, the expression of a sugar efflux transporter from *Yersinia* was shown to enable the export of the human milk oligosaccharide 2'-fucosyllactose out of an engineered *E. coli* production strain.

(13) Apart from this, from a mechanistic and energetic point of view, only the ion-gradient-driven transport systems have the potential to translocate solutes in both directions across the membrane. This is exemplarily true for the above mentioned LacY, a galactoside/H.sup.+ symporter, which is part of the bacterial lac operon that allows the metabolism of lactose in *E. coli*. This permease primarily imports lactose into the cell but it is also capable to transfer its substrate in the opposite direction.

(14) Besides the major facilitator superfamily, which represents the largest group of transporters, bacteria possess further mechanisms to excrete solutes—often summarized in the classes of multidrug efflux pumps. Alike for the MFS, the activities of the small multidrug resistance superfamily (SMR), the multidrug and toxic compound extrusion superfamily (MATE) and the resistance-nodulation-cell division superfamily (RND) rely on the electrochemical gradient. The fifth class is the adenosine triphosphate (ATP)-binding cassette superfamily (ABC) which uses ATP as energy source to drive molecules from the cell. As for the MFS, members of SMR, MATE, RND and ABC transport structurally diverse molecules. Further, most of their so far identified substrates are not naturally occurring, and, thus, their preferences are hardly predictable.

(15) Although chemical synthesizing processes are known for human milk oligosaccharides, these

processes are very cost-intensive and do not lead to satisfying amounts. On the other hand, fermentation processes using genetically modified microorganisms still have the drawback that the export of larger oligosaccharides (tetra-, penta-, hexasaccharides) represents a major limitation for the establishment of cost effective production processes. As a consequence, there still is the need for improved processes for the production of large-scale human oligosaccharides.

SUMMARY

(16) According to the invention, this and other objects are solved by the methods and microbial host cell(s) as claimed in the attached claims.

(17) With the methods and host cells according to the invention it is possible to produce a desired oligosaccharide, preferably an oligosaccharide that is not produced in an unmodified host cell, and also preferably an oligosaccharide belonging to the human milk oligosaccharides, in large amounts obtainable from the medium. As such, the oligosaccharide is, so to say, obtainable in free form in the medium; it is not bound to a surface protein or membrane protein or other protein of the surface of the host cell.

(18) According to the invention, a method for the production of a desired oligosaccharide by a genetically modified microbial host cell, comprising the steps of a) providing a genetically modified microbial host cell that comprises at least one recombinant glycosyltransferase, and that has the expression or activity of at least one endogenous sugar export protein modified such, that the expression or activity of the sugar export protein is either (i) increased or (ii) decreased or inactivated as compared to an genetically unmodified host cell, so that (i) the export of a oligosaccharide into the medium is either decreased or abolished, or (ii) the transport of a desired oligosaccharide is increased, respectively, as compared to an genetically unmodified host cell, b) cultivating the host cell in a medium under conditions permissive for the production of the desired oligosaccharide, whereby the desired oligosaccharide is transported into the medium. The method may further comprise the step of c) obtaining the desired oligosaccharide from the medium.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 shows a schematic illustration for the production of either lacto-N-triose II or lacto-N-tetraose in a host cell cultivated in a medium;

(2) FIG. 2 shows the results of the TLC analysis of culture extracts of lacto-N-triose II (LNT II) producing *E. coli* BL21(DE3) strains overexpressing the β -1,3-N-acetyl glucosaminyltransferase gene PmnagT(13, 14);

(3) FIG. 3 shows the results of the TLC analysis of culture extracts of lacto-N-tetraose (LNT) producing *E. coli* BL21(DE3) strains overexpressing the β -1,4-galactosyltransferase encoding genes BfgalT2 (1), PmgalT7 (3), MsgalT8 (6), gatD (7), lex1 (9), IgtB (11) or IsgD (13);

(4) FIG. 4 shows the results of the TLC analysis of culture extracts of lacto-N-tetraose (LNT) producing *E. coli* BL21(DE3) strains overexpressing the β -1,4-galactosyltransferase encoding genes KdgalT10 (1), cpsI14J (7), cpsIaJ (8, 9), HpgalT (12);

(5) FIG. 5 shows the results of the TLC analysis of culture extracts of lacto-N-tetraose producing *E. coli* BL21(DE3) strains overexpressing the β -1,4-galactosyltransferase encoding gene waaX (5);

(6) FIG. 6 shows the results of the TLC analysis of culture extracts of lacto-N-tetraose producing *E. coli* BL21(DE3) strains overexpressing the β -1,3-galactosyltransferase encoding genes wbdO or furA;

(7) FIG. 7 shows the results of HPLC analyses of the culture supernatant of lacto-N-tetraose producing *E. coli* BL21 (DE3) strain. (A) Supernatant of *E. coli* BL21(DE3) 1353 and 1431 grown in the presence of glucose and lactose after 24 h of incubation. (B) Supernatant of *E. coli* BL21(DE3) 1353 and 1431 grown in the presence of glucose and lactose after 48 h of incubation;

(8) FIG. 8 shows a diagram depicting the relative concentration of lacto-N-tetraose in the supernatant of *E. coli* BL21 (DE3) strains overexpressing sugar efflux transporters compared to the control strain 1353; and

(9) FIG. 9 shows a diagram depicting concentrations of lacto-N-triose II in the supernatant of *E. coli* BL21 (DE3) strains overexpressing the sugar efflux transporters TP11 (2), YjhB (3) or TP70 (4).

DETAILED DESCRIPTION OF A PREFERRED EMBODIMENT

(10) In the method according to the invention, it is preferred if the desired oligosaccharide is a human milk oligosaccharide comprising a lacto-N-triose II (LNT-II; GlcNAc(β 1-3)Gal(β 1-4)Gluc) as a core trisaccharide. In this connection, an oligosaccharide having a “core trisaccharide” is meant to comprise the specific trisaccharide representing the reducing end of a desired oligosaccharide, and comprising, as the case may be, additional saccharide moieties, with the specific trisaccharide representing the major moiety.

(11) Accordingly, in an embodiment of the method and the host cell according to the invention, the desired oligosaccharide is selected from the group consisting of: lacto-N-triose II, lacto-N-tetraose, lacto-N-neotetraose, lacto-N-fucopentaose I, lacto-N-fucopentaose II, lacto-N-fucopentaose III, lacto-N-fucopentaose V, lacto-N-difucosylhexose I, lacto-N-difucosylhexaose II, lacto-N-sialylpentaose LSTa, LSTb, LSTc, disialyllacto-N-tetraose, disialyllacto-N-neotetraose.

(12) In order to overcome the above mentioned drawbacks of limited oligosaccharide export the object is further solved by a method according to the invention, wherein the host cell comprises: at least one homologous or heterologous nucleic acid sequence coding for a protein enabling the export of a desired oligosaccharide into the culture medium, wherein said host cell has been modified such, that the expression of the homologous or heterologous nucleic acid sequence is overexpressed or under control of a promoter enabling the overexpression of the nucleic acid sequence; and/or the deletion, disruption, diminishment or inactivation of at least one endogenous nucleic acid sequence coding for an exporter protein that exports precursors of the desired oligosaccharide outside the host cell; and/or at least one homologous or heterologous nucleic acid sequence coding for a protein mediating the import of a precursor of a desired oligosaccharide into said host cell, wherein preferably the nucleic acid sequence is overexpressed, and wherein preferably the precursor is larger than a disaccharide.

(13) The genetically modified microbial host cell comprising the characteristics as set forth herein are cultured in the presence of glucose, sucrose, glycerin or a combination thereof—using these substrates as carbon- and energy sources—as well as in the presence of lactose or oligosaccharides larger than disaccharides, e.g., LNT-II.

(14) In a preferred embodiment of this method and host cell, said protein enabling the export of a desired oligosaccharide belongs to the class of secondary active transporters, and more preferably effects the export of an oligosaccharide comprising at least three moieties.

(15) According to preferred embodiments, for the export of desired oligosaccharides a suitable exporter is expressed in addition to the genes that are responsible for intracellular oligosaccharide biosynthesis.

(16) According to one aspect of the method and host cell of the invention, the at least one nucleic acid sequence coding for a protein enabling the export of a desired oligosaccharide is an endogenous or a recombinant nucleic acid.

(17) In a preferred embodiment of the method and the host cell of the invention, the nucleic acid sequence coding for a protein enabling the export of a desired oligosaccharide is of bacterial, archeal, plant, yeast or animal origin; preferably, the at least one nucleic acid sequence coding for a protein enabling the export of a desired oligosaccharide is a gene selected from the group consisting of yebQ and yjhB from *Escherichia coli*, proP from *Mannheimia succiniciproducens* and setA from *Cedecea neteri* or functional fragments thereof.

(18) Preferably, the oligosaccharide exporter is a protein selected from at least one of the following:

SetA, SetB, SetC, YdeA, Cmr, YnfM, MdtD, YfhC, YhhS, YhdA, YdeE, MhpT, YebQ, YjhB, Bcr and YdeA of *E. coli*, or ProP from *Mannheimia succiniciproducens* and SetA from *Cedecea neteri* or variants or homologs thereof.

(19) In yet another preferred embodiment, the recombinant glycosyltransferase is selected from at least one of the following: a galactosyltransferase, a sialyltransferase, an N-acetylglucosaminyltransferase and a fucosyltransferase, and is preferably selected from at least one of the following: β -1,3-N-acetylglucosaminyltransferase, β -1,3-galactosyltransferase, β -1,4-galactosyltransferase, β -1,6-galactosyltransferase, α -2,3-sialyltransferase, α -2,6-sialyltransferase, α -1,2-fucosyltransferase, or α -1,3-fucosyltransferase.

(20) A preferred embodiment of the method and the host cell of the invention, concerns the a host cell or its provision, wherein the host cell comprises (i) a β -1,3-N-acetylglucosaminyltransferase, and (ii) a β -1,3-galactosyltransferase or a β -1,4-galactosyltransferase as glycosyltransferases. In this connection it is preferred, if said β -1,3-N-acetylglucosaminyltransferase has the activity of ligating N-acetylglucosamine to lactose generating lacto-N-triose II, and if said β -1,3-galactosyltransferase or said β -1,4-galactosyltransferase, respectively, have the activity to galactosylate lacto-N-triose II thus generating lacto-N-tetraose or lacto-N-neotetraose, respectively. The here developed system is easily adaptable to even more complex oligosaccharides by the expression of further glycosyltransferases.

(21) With the microbial cell and the method according to the invention, it is possible to ferment a desired oligosaccharide in large quantities, especially an oligosaccharide comprising LNT-II as core structure, and to recover it from the culture broth.

(22) In a preferred embodiment, said β -1,3-N-acetylglucosaminyltransferase belongs to the class of lgtA of *Neisseria meningitides* or PmnagT of *Pasteurella multocida*, or variants thereof.

(23) Preferably, the glycosyltransferase is selected from a galactosyltransferase, a sialyltransferase, an N-acetylglucosaminyltransferase and a fucosyltransferase.

(24) In yet another preferred embodiment, the lacto-N-tetraose generating β -1,3-galactosyltransferase is WbdO or a functional variant thereof. According to an aspect of the invention, the β -1,3-galactosyltransferase is a β -1,3-galactosyltransferase derived from *Salmonella enterica* (wbdO, acc. no. AY730594), and is preferably encoded by a gene selected from the group consisting of wbgO from *Escherichia coli* O55:H7 or furA from *Lutiella nitroferum*, or a functional fragments thereof.

(25) The invention also concerns a genetically modified microbial host cell, preferably a bacterial host cell, as described above in which the endogenous β -galactosidase gene is inactivated or deleted and in which a functional lactose permease gene is present.

(26) Accordingly, in a preferred embodiment of the method and the host cell of the invention, a genetically modified host cell is provided, in which, where applicable, an endogenous β -galactosidase gene and a glucosamine-6-phosphate deaminase gene are inactivated or deleted, and wherein said genetically modified host cell comprises a nucleic acid sequence coding for a functional lactose permease protein, preferably LacY.

(27) In a preferred embodiment, the genetically modified host cell comprises an increased UDP-N-acetylglucosamine and UDP-galactose, GDP-fucose or CMP-N-acetylneuraminic acid production capability as compared to a genetically unmodified host cell.

(28) In a refinement of this embodiment of the method of and of the host cell of the invention, said increased UDP-N-acetylglucosamine and UDP-galactose production capability comprises the overexpression of one or more genes encoding for proteins comprising the following activities for a: L-glutamine: D-fructose-6-phosphate aminotransferase, N-acetyl glucosamine-1-phosphate uridylyltransferase/glucosamine-1-phosphate acetyl transferase, phosphoglucosamine mutase, UDP-galactose-4-epimerase, phosphoglucomutase, glucose-1-phosphate uridylyltransferase.

(29) For the synthesis of, e.g. LNT, UDP-galactose and UDP-N-acetylglucosamine are required. UDP-galactose can be obtained by feeding galactose to the HMO producing bacterial host cell via

the fermentation medium. The galactose is then taken up by the cell, phosphorylated to galactose-1-phosphate and then converted to UDP-galactose. Genes encoding these enzymatic activities are well known in the literature (Grossiord et al., J. Bacteriol 2003 185(3) 870-878). The supply for UDP-galactose can be also obtained from the cells own metabolism, and the metabolism can be improved by further genetic modification, such as the overexpression of the UDP-galactose-4'-epimerase, or the UDP-galactose-4'-epimerase in combination with the glucose-1-phosphate-1-uridyltransferase. UDP-N-acetylglucosamine can be also obtained from the bacterial host cell's own UDP-N-acetylglucosamine metabolism. The provision of UDP-N-acetylglucosamine for the synthesis of N-acetylglucosamine containing oligosaccharides can be improved by the inactivation of the N-acetylglucosamine catabolism within the producing cell.

(30) According to one aspect of the invention, the genetically modified host cell is cultivated in the presence of glucose, sucrose, glycerol or a combination thereof, but neither by addition or in the presence of N-acetylglucosamine or galactose nor in a combination thereof.

(31) In a preferred embodiment of the method and of the host cell of the invention, the desired oligosaccharide is lacto-N-triose II, which is produced by total fermentation from a simple carbon source in the host cell by the action of the heterologous expressed glycosyltransferases β -1,4-galactosyltransferase and β -1,3-N-acetylglucosaminyltransferase.

(32) The present invention, as already mentioned above, also concerns a genetically modified host cell for the production of a desired oligosaccharide, the oligosaccharide comprising a lacto-N-triose II (LNT-II; GlcNAc(β 1-3)Gal(β 1-4)Gluc) as a core trisaccharid, wherein the host cell comprises at least one recombinant glycosyltransferase, the glycosyltransferase being preferably selected from a galactosyltransferase, a sialyltransferase, and an N-acetylglucosaminyltransferase, and has the expression or activity of at least endogenous sugar transport protein modified such, that the expression or activity of the endogenous sugar transport protein is functionally inactivated for the export of a precursor of the desired oligosaccharide.

(33) A preferred embodiment concerns a host cell as described above, comprising (i) a heterologous expressed β -1,3-N-acetylglucosaminyltransferase, and (ii) a heterologous expressed β -1,3-galactosyltransferase or a heterologous expressed β -1,4-galactosyltransferase as glycosyltransferases, wherein the host cell further preferably comprises at least one homologous or heterologous nucleic acid sequence coding for a protein enabling the export of the oligosaccharide into a culture medium the host cell is cultivated in, wherein said protein enabling the export of the desired oligosaccharide belongs to the class of secondary active transporters, wherein said host cell has been modified such, that the expression of the homologous or heterologous nucleic acid sequence is overexpressed or under control of a promoter enabling the overexpression of the nucleic acid sequence. In preferred embodiments of the host cell, said at least one nucleic acid sequence coding for a protein enabling the export of the desired oligosaccharide is an endogenous or a recombinant nucleic acid sequence.

(34) As already outlined for the method according to the invention, it is also preferred in the host cell of the invention, if said nucleic acid sequence coding for a protein enabling the export of a desired oligosaccharide is of bacterial, archeal, plant, yeast or animal origin.

(35) According to another aspect of the invention, the host cell as described above further comprises: the deletion, disruption, diminishment or inactivation of at least one endogenous nucleic acid sequence coding for an exporter protein that exports precursors of the desired oligosaccharide outside the host cell; and/or at least one homologous or heterologous nucleic acid sequence coding for a protein enabling the import of a precursor of a desired oligosaccharide into said host cell, wherein preferably the nucleic acid sequence is overexpressed, and wherein preferably the precursor is larger than a disaccharide.

(36) With the overexpression of at least one homologous or heterologous nucleic acid sequence coding for a protein enabling the import of a precursor of a desired oligosaccharide into said host cell, it is possible to feed precursors of a desired oligosaccharide to the culture medium, which get

imported into the host cell, such as, e.g., LNT-II.

(37) According to one aspect of the invention, in the host cell said at least one nucleic acid sequence coding for a protein enabling the export of a desired oligosaccharide is a gene selected from the group consisting of yebQ and yjhB from *Escherichia coli*, proP from *Mannheimia succiniciproducens* and setA from *Cedecea neteri* or functional fragments thereof.

(38) According to yet another preferred embodiment, the desired oligosaccharide is lacto-N-triose II, and the protein enabling the export of the oligosaccharide into a culture medium the host cell is cultivated in, is YjhB from *Escherichia coli*, ProP from *Mannheimia succiniciproducens* and SetA from *Cedecea neteri* or functional fragments thereof.

(39) According to a preferred embodiment, the microbial host according to the invention is further modified not to express proteins exporting precursors of a desired oligosaccharide.

(40) In a preferred embodiment of the host cell, the desired oligosaccharide is lacto-N-tetraose, the precursor is lacto-N-triose II, and the host cell has deleted, disrupted or inactivated at least one nucleic acid sequence coding for an exporter protein that is able to export lacto-N-triose II outside the host cell.

(41) In this connection it is preferred, if the protein enabling the export of lacto-N-tetraose is selected from YebQ from *Escherichia coli* BL21(DE3), SpoVB of *Bacillus amyloliquefaciens*, YabM of *Erwinia pyrrifolia*, Bcr of *E. coli* MG1655, YdeA of *E. coli* MG1655, ProP2 of *Haemophilus parainfluenzae*, SetA of *Pectobacterium carotovorum*, FucP of *E. coli* MG1655, MdeA of *Staphylococcus aureus* Bmb9393, ImrA of *Lactococcus lactis*, SetA of *Pseudomonas* sp. MT-1 and SetA of *Beauveria bassiana* D1-5.

(42) Preferably, the oligosaccharide exporter is a protein selected from at least one of the following: SetA, SetB, SetC, YdeA, Cmr, YnfM, MdtD, YfcJ, YhhS, EmrD, YdhC, YbdA, YdeE, MhpT, YebQ, YjhB, Bcr and YdeA of *E. coli*, or ProP from *Mannheimia succiniciproducens* and SetA from *Cedecea neteri* or variants or homologs thereof.

(43) Presently, the term “nucleic acid” refers to a single- or double-stranded deoxyribonucleotide or ribonucleotide macromolecule and encompasses known analogues or natural or synthetically produced nucleotides that hybridize with the desired nucleic acid and that encode a certain polypeptide.

(44) The term “recombinant” or “genetically modified”, as used herein with reference to a microbial host cell indicates that the microbial host cell replicates a heterologous nucleic acid, or expresses a peptide or protein encoded by a heterologous nucleic acid (i.e., a sequence “foreign to said cell”). Recombinant cells can contain genes that are not found within the native (non-recombinant) form of the cell. Recombinant cells can also contain genes found in the native form of the cell wherein the genes are modified and re-introduced into the cell by artificial means. The term also encompasses cells that contain a nucleic acid endogenous to the cell that has been modified without removing the nucleic acid from the cell; such modifications include those obtained by gene replacement, site-specific mutation, and related techniques. Accordingly, a “recombinant polypeptide” is one which has been produced by a recombinant cell. A “heterologous sequence” or a “heterologous nucleic acid”, as used herein, is one that originates from a source foreign to the particular host cell (e.g. from a different species), or, if from the same source, is modified from its original form. Thus, a heterologous nucleic acid operably linked to a promoter is from a source different from that from which the promoter was derived, or, if from the same source, is modified from its original form. The heterologous sequence may be stably introduced, e.g. by transfection, transformation, conjugation or transduction, into the genome of the host microbial host cell, thus representing a genetically modified host cell. Techniques may be applied which will depend on the host cell the sequence is to be introduced. Various techniques are known to a person skilled in the art and are, e.g., disclosed in Sambrook et al., *Molecular Cloning: A Laboratory Manual*, 2nd Ed., Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y. (1989).

(45) Accordingly, a “microbial host cell” is presently understood as a microbial, preferably

bacterial, cell which has been transformed or transfected, or is capable of transformation or transfection by an exogenous polynucleotide sequence.

(46) Thus, the nucleic acid sequences as used in the present invention, may, e.g., be comprised in a vector which is to be stably transformed/transfected or otherwise introduced into host microorganism cells.

(47) Presently, the term “operably linked” as used herein, shall mean a functional linkage between a nucleic acid expression control sequence (such as a promoter, signal sequence, or array of transcription factor binding sites) and a second nucleic acid sequence, wherein the expression control sequence affects transcription and/or translation of the nucleic acid corresponding to the second sequence. Accordingly, the term “Promoter” designates DNA sequences which usually “precede” a gene in a DNA polymer and provide a site for initiation of the transcription into mRNA. “Regulator” DNA sequences, also usually “upstream” of (i.e., preceding) a gene in a given DNA polymer, bind proteins that determine the frequency (or rate) of transcriptional initiation. Collectively referred to as “promoter/regulator” or “control” DNA sequence, these sequences which precede a selected gene (or series of genes) in a functional DNA polymer cooperate to determine whether the transcription (and eventual expression) of a gene will occur. DNA sequences which “follow” a gene in a DNA polymer and provide a signal for termination of the transcription into mRNA are referred to as transcription “terminator” sequences.

(48) A great variety of expression systems can be used to produce the polypeptides of the invention. Such vectors include, among others, chromosomal, episomal and virus-derived vectors, e.g., vectors derived from bacterial plasmids, from bacteriophage, from transposons, from yeast episomes, from insertion elements, from yeast chromosomal elements, from viruses, and vectors derived from combinations thereof, such as those derived from plasmid and bacteriophage genetic elements, such as cosmids and phagemids. The expression system constructs may contain control regions that regulate as well as engender expression. Generally, any system or vector suitable to maintain, propagate or express polynucleotides and to synthesize a polypeptide in a host may be used for expression in this regard. The appropriate DNA sequence may be inserted into the expression system by any of a variety of well-known and routine techniques, such as, for example, those set forth in Sambrook et al., supra.

(49) The art is rich in patent and literature publications relating to “recombinant DNA” methodologies for the isolation, synthesis, purification and amplification of genetic materials for use in the transformation of selected host organisms. Thus, it is common knowledge to transform host organisms with “hybrid” viral or circular plasmid DNA which includes selected exogenous (i.e. foreign or “heterologous”) DNA sequences. The procedures known in the art first involve generation of a transformation vector by enzymatically cleaving circular viral or plasmid DNA to form linear DNA strands. Selected foreign DNA strands usually including sequences coding for desired protein product are prepared in linear form through use of the same/similar enzymes. The linear viral or plasmid DNA is incubated with the foreign DNA in the presence of ligating enzymes capable of effecting a restoration process and “hybrid” vectors are formed which include the selected exogenous DNA segment “spliced” into the viral or circular DNA plasmid.

(50) As used herein, the term “cultivating” means growing a bacterial cell in a medium and under conditions permissive and suitable for the production of the desired oligosaccharide(s). A couple of suitable bacterial host cells as well as mediums and conditions for their cultivation will be readily available for one skilled in the art upon reading the disclosure of this invention in connection with the skilled person's technical and expert background.

(51) As used herein, the term “recovering” or “obtaining” means isolating, harvesting, purifying, collecting or otherwise separating from the host cell culture the oligosaccharide produced by the host cell according to the invention.

(52) A “microbial” host cell according to the invention, and as generally understood, means any microorganism, including bacteria, fungi and archaea, which is generally suitable for cultivation in

large amounts, and which can be genetically modified according to the invention in order to produce a desired oligosaccharide. Preferred microorganisms are bacteria, e.g. *Escherichia coli*, *Corynebacterium glutamicum* and the yeast *Saccharomyces* sp., which have the advantage that these microorganisms can be grown easily and inexpensively in laboratory settings, and the bacteria and yeast have been intensively investigated for over many years

(53) Generally, and throughout the present invention, the term “glycosyltransferase activity” or “glycosyltransferase” designates and encompasses enzymes that are responsible for the biosynthesis of disaccharides, oligosaccharides and polysaccharides, and they catalyze the transfer of monosaccharide moieties from an activated nucleotide monosaccharide/sugar (the “glycosyl donor”) to a glycosyl acceptor molecule.

(54) Generally, and throughout the present invention, the terms “exporter” or “exporter protein” or “protein enabling the export of a desired oligosaccharide”, which terms are presently being used synonymously, designates one or more polypeptides that solely or as part of a multi-protein complex transfers an oligosaccharide from the intracellular milieu of a bacterial cell into the periplasm of said cell or the culture supernatant, thus, enabling the oligosaccharide to pass the cellular membrane and/or the cell wall of said cell.

(55) Within the scope of the present invention, also nucleic acid/polynucleotide and polypeptide polymorphic variants, alleles, mutants, and interspecies homologs are comprised by those terms, that have an amino acid sequence that has greater than about 60% amino acid sequence identity, 65%, 70%, 75%, 80%, 85%, 90%, preferably 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% or greater amino acid sequence identity, preferably over a region of at least about 25, 50, 100, 200, 500, 1000, or more amino acids, to a polypeptide encoded by a wild type glycosyltransferase activity or oligosaccharide export displaying protein.

(56) “Variant(s)” as the term is used herein, is a polynucleotide or polypeptide that differs from a reference polynucleotide or polypeptide, respectively, but retains essential properties. A typical variant of a polynucleotide differs in nucleotide sequence from another, reference polynucleotide. Changes in the nucleotide sequence of the variant may or may not alter the amino acid sequence of a polypeptide encoded by the reference polynucleotide. Nucleotide changes may result in amino acid substitutions, additions, deletions, fusions and truncations in the polypeptide encoded by the reference sequence, as discussed below. A typical variant of a poly-peptide differs in amino acid sequence from another, reference polypeptide. Generally, differences are limited so that the sequences of the reference polypeptide and the variant are closely similar overall and, in many regions, identical. A variant and reference polypeptide may differ in amino acid sequence by one or more substitutions, additions, deletions in any combination. A substituted or inserted amino acid residue may or may not be one encoded by the genetic code. A variant of a polynucleotide or polypeptide may be a naturally occurring such as an allelic variant, or it may be a variant that is not known to occur naturally. Non-naturally occurring variants of polynucleotides and polypeptides may be made by mutagenesis techniques, by direct synthesis, and by other recombinant methods known to the persons skilled in the art.

(57) Accordingly, a “functional fragment” of any of the genes/proteins disclosed therein, is meant to designate sequence variants of the genes/proteins still retaining the same or somewhat lesser activity of the gene or protein the respective fragment is derived from.

(58) In this connection, the term “nucleic acid sequence encoding . . .” generally refers to any polyribonucleotide or polydeoxyribonucleotide, which may be unmodified RNA or DNA or modified RNA or DNA, and generally represents a gene which encodes a certain polypeptide or protein.

(59) In this context, the term “polypeptide(s)” refers to any peptide or protein comprising two or more amino acids joined to each other by peptide bonds or modified peptide bonds.

“Polypeptide(s)” refers to both short chains, commonly referred to as peptides, oligopeptides and oligomers and to longer chains generally referred to as “proteins”. Polypeptides may contain amino

acids other than the 20 gene encoded amino acids. "Polypeptide(s)" include those modified either by natural processes, such as processing and other post-translational modifications, but also by chemical modification techniques. It will be appreciated that the same type of modification may be present in the same or varying degree at several sites in a given polypeptide, without essentially altering the activity of the polypeptide. Also, a given polypeptide may contain many types of modifications. Modifications can occur anywhere in a polypeptide, including the peptide backbone, the amino acid side-chains, and the amino or carboxyl termini.

(60) Further, with the expression "precursor" compounds are encompassed which are involved in the biosynthetic pathway of the oligosaccharide according to the invention or which are produced and naturally present in the host cell.

(61) A "precursor that is larger than a disaccharide" is presently understood as a sugar moiety that comprises more than two monosaccharide residues.

(62) The term "desired oligosaccharide" refers to a sugar polymer consisting of at least three moieties, thus, comprising trisaccharides, tetrasaccharides, pentasaccharides etc., preferably an oligosaccharide selected from at least one of the following: lacto-N-triose II, lacto-N-tetraose, lacto-N-neotetraose, lacto-N-fucopentaose I, lacto-N-fucopentaose II, lacto-N-fucopentaose III, lacto-N-fucopentaose V, lacto-N-difucosylhexose I, lacto-N-difucosylhexaose II, lacto-N-sialylpentaose LSTa, LSTb, LSTc, disialyllacto-N-tetraose, disialyllacto-N-neotetraose.

(63) Presently, and as generally understood in the relevant field, the expression "homologous" refers to a nucleic acid sequence that encodes for a specific product or products and is derived from the same species, in which said nucleic acid sequence is inserted. Accordingly, the term "heterologous" refers to a nucleic acid sequence encoding for a specific product or products and being derived from a species other than those in which said nucleic acid sequence is inserted.

(64) The term "endogenous" herein and generally within the field means that the nucleic acid encoding for an enzyme of interest is originating from the bacterial host cell and has not been introduced into said host cell, whereas a "recombinant" nucleic acid has been introduced into said host cell and does not originate from said host cell.

(65) The expression "overexpressed", or "overexpressing" or "under control of a promoter sequence enabling the overexpression of said nucleic acid sequence" presently, and generally in the art, means the expression of a gene in greater-than-normal amounts, i.e. in increased quantity thus leading to an increased amount of the protein the nucleic acid sequence is coding for.

(66) In some embodiments, the nucleic acid sequence is placed under the control of an inducible promoter, which is a promoter that directs expression of a gene where the level of expression is alterable by environmental or developmental factors such as, for example, temperature, pH, anaerobic or aerobic conditions, light, transcription factors and chemicals. Such promoters are referred to herein as "inducible" promoters, which allow one to control the timing of expression of the proteins used in the present invention. For *E. coli*, and other microbial host cells, inducible promoters are known to those of skill in the art.

(67) Further advantages are evident from the description and the drawings.

(68) It is understood that the features mentioned above and those yet to be explained below can be used not only in the respective combinations indicated, but also in other combinations or in isolation, without leaving the context of the present invention.

(69) The invention will be described in more detail in the examples and the attached figures, in which

(70) FIG. 1 shows a schematic illustration for the production of either lacto-N-triose II or lacto-N-tetraose in a host cell cultivated in a medium;

(71) FIG. 2 shows the results of the TLC analysis of culture extracts of lacto-N-triose II (LNT II) producing *E. coli* BL21(DE3) strains overexpressing the β -1,3-N-acetyl glucosaminyltransferase gene PmnagT(13, 14);

(72) FIG. 3 shows the results of the TLC analysis of culture extracts of lacto-N-tetraose (LNT)

producing *E. coli* BL21(DE3) strains overexpressing the β -1,4-galactosyltransferase encoding genes BfgalT2 (1), PmgalT7 (3), MsgalT8 (6), gatD (7), lex1 (9), IgtB (11) or IsgD (13); (73) FIG. 4 shows the results of the TLC analysis of culture extracts of lacto-N-tetraose (LNT) producing *E. coli* BL21(DE3) strains overexpressing the β -1,4-galactosyltransferase encoding genes KdgalT10 (1), cpsI14J (7), cpslaJ (8, 9), HpgalT (12); (74) FIG. 5 shows the results of the TLC analysis of culture extracts of lacto-N-tetraose producing *E. coli* BL21(DE3) strains overexpressing the β -1,4-galactosyltransferase encoding gene waaX (5); (75) FIG. 6 shows the results of the TLC analysis of culture extracts of lacto-N-tetraose producing *E. coli* BL21(DE3) strains overexpressing the β -1,3-galactosyltransferase encoding genes wbdO or furA; (76) FIG. 7 shows the results of HPLC analyses of the culture supernatant of lacto-N-tetraose producing *E. coli* BL21 (DE3) strain. (A) Supernatant of *E. coli* BL21(DE3) 1353 and 1431 grown in the presence of glucose and lactose after 24 h of incubation. (B) Supernatant of *E. coli* BL21(DE3) 1353 and 1431 grown in the presence of glucose and lactose after 48 h of incubation; (77) FIG. 8 shows a diagram depicting the relative concentration of lacto-N-tetraose in the supernatant of *E. coli* BL21 (DE3) strains overexpressing sugar efflux transporters compared to the control strain 1353; and (78) FIG. 9 shows a diagram depicting concentrations of lacto-N-triose II in the supernatant of *E. coli* BL21 (DE3) strains overexpressing the sugar efflux transporters TP11 (2), YjhB (3) or TP70 (4).

EXAMPLES

(79) FIG. 1 shows a schematic drawing of an exemplary host cell 10 according to the invention, importing lactose and synthesizing lacto-N-triose II (LNT 11) and lacto-N-tetraose (LNT). Lactose is imported from the medium the host cell is cultivated in into the cell via transporter 1. The enzyme N-acetylglucosaminyltransferase NacGlcT ligates N-acetylglucosamine to the acceptor substrate lactose, thus generating LNT-II. LNT-II is exported from the cell via exporter protein 20. Since LNT-II is a precursor of LNT or LNnT, the exporter exporting LNT-II represents an exporter protein exporting precursors of the latter oligosaccharides. As can further be seen from FIG. 1, the cell comprises a protein having β -1,3-galactosyltransferase activity enabling the galactosylation of LNT-II to intracellularly generate LNT; the cell may also and/or alternatively comprise or β -1,4-galactosyltransferase activity enabling the galactosylation of LNT-II to intracellularly generate lacto-N-neotetraose LNnt. LNT—or as the case may be LNnt—is then exported, via a oligosaccharide exporter from the cell into the culture medium the cell is cultivated in. (80) The exporters are membrane-bound, and their expression can be either overexpressed, which—in case of overexpression of the LNT-II exporter leads to an increased LNT-II export and to a decreased LNT export, whereas when the LNT-II exporting exporter protein is deleted or otherwise inactivated, this leads to an improved LNT-export. The LNT-II exporter preferably is an endogenous exporter protein, whereas the LNT-exporter protein preferably is a heterologous exporter protein.

Example 1

(81) Development of an *E. coli* Lacto-N-Triose II Production Strain

(82) *Escherichia coli* BL21(DE3) was used to construct a lacto-N-triose II (LNT-2) producing strain. Metabolic engineering included mutagenesis and deletions of specific genes, respectively, and genomic integrations of heterologous genes. The genes lacZ and araA were inactivated by mutagenesis using mismatch-oligonucleotides as described by Ellis et al., “High efficiency mutagenesis, repair, and engineering of chromosomal DNA using single-stranded oligonucleotides”, Proc. Natl. Acad. Sci. USA 98: 6742-6746 (2001).

(83) Genomic deletions were performed according to the method of Datsenko and Warner (Proc. Natl. Acad. Sci. USA 97:6640-6645 (2000)). To prevent intracellular degradation of N-acetylglucosamine, genes encoding N-acetylglucosamine-6-phosphate deacetylase (nagA) and

glucosamine-6-phosphate deaminase (nagB) were deleted from the genome of the *E. coli* strain BL21 (DE3) strain. Also genes *wzxC-wcaJ* were deleted. WcaJ encodes an UDP-glucose:undecaprenyl phosphate glucose-1-phosphate transferase catalysing the first step in colanic acid synthesis (Stevenson et al., J. Bacteriol. 1996, 178:4885-4893). In addition the genes *fucI* and *fucK*, coding for L-fucose isomerase and L-fuculose kinase, respectively, were removed.

(84) Genomic integration of heterologous genes was performed by transposition. Either the EZ-Tn5™ transposase (Epicentre, USA) was used to integrate linear DNA-fragments or the hyperactive C9-mutant of the mariner transposase Himarl (Lampe et al., Proc. Natl. Acad. Sci. 1999, USA 96:11428-11433) was employed for transposition. To produce EZ-Tn5 transposomes the gene of interest together with a FRT-site flanked antibiotic resistance marker was amplified with primer 1119 and 1120 (all primer used are listed in table 3 below); the resulting PCR-product carried on both sites the 19-bp Mosaic End recognition sites for the EZ-Tn5 transposase. For integration using Himarl transposase expression constructs (operons) of interest were similarly cloned together with a FRT-site flanked antibiotic resistance marker into the pEcomar vector. The pEcomar vector encodes the hyperactive C9-mutant of the mariner transposase Himarl under the control of the arabinose inducible promoter P.sub.araB. The expression fragment <P.sub.tet-lacY-FRT-aadA-FRT> (SeqID1) was integrated by using the EZ-Tn5 transposase. After successful integration of the gene for the lactose importer LacY from *E. coli* K12 TG1 (acc. no. ABN72583) the resistance gene was eliminated from streptomycin resistant clones by the FLP recombinase encoded on plasmid pCP20 (Datsenko and Warner, Proc. Natl. Acad. Sci. 2000, USA 97:6640-6645). The N-acetylglucosaminyltransferase gene *lgtA* from *Neisseria meningitidis* MC58 (acc. no. NP_274923) was codon-optimized for expression in *E. coli* and prepared synthetically by gene synthesis. Together with the gene *galT*, encoding a galactose-1-phosphate uridylyltransferase from *E. coli* K-12 substr. MG1655 (acc. no. NP_415279) that was similarly obtained by gene synthesis, *lgtA* was inserted by transposition (SeqID2) using plasmid pEcomar-*lgtA-galT*. To enhance de novo synthesis of UDP-N-acetylglucosamine, genes encoding L-glutamine:D-fuctose-6-phosphate aminotransferase (*glmS*), phosphoglucosamine mutase from *E. coli* K-12 substr. MG1655 (*glmM*) and N-acetylglucosamine-1-phosphate uridylyltransferase/glucosamine-1-phosphate acetyltransferase (*glmU*) from *E. coli* K-12 substr. MG1655 (acc. no. NP_418185, NP_417643, NP_418186, respectively) were codon-optimized and obtained by gene synthesis. The operon *glmUM* was cloned under the control of constitutive tetracyclin promoter P.sub.tet while *glmS* was cloned under the constitutive P.sub.T5 promoter. The transposon cassette <P.sub.tet-*glmUM*-P.sub.T5-*glmS*-FRT-dhfr-FRT> (SeqID3), flanked by the inverted terminal repeats specifically recognized by the mariner-like element Himarl transposase was inserted from pEcomar-*glmUM-glmS* revealing a lacto-N-triose II production strain. Additionally, the expression fragment <P.sub.tet-lacY(6H/S)-FRT-aadA-FRT> (SeqID4) was integrated by using the EZ-Tn5 transposase.

(85) The *gal*-operon (*galETKM*) was amplified from *E. coli* K12 TG1 (SeqID6) using primer 605 and 606 and inserted into the *galM ybhJ* locus of *E. coli* BL21 (DE3) strain by homologous recombination facilitated by using the red recombinase helper plasmid pKD46 (Datsenko and Warner, Proc. Natl. Acad. Sci. 2000, USA 97:6640-6645). Sequences of the heterologous genes and gene clusters are deposit in appendix 1.

Example 2

(86) Batch Fermentation of *E. coli* BL21 (DE3) 707 screening various β -1,3-N-acetyl-glycosaminyltransferases

(87) The gene for the β -1,3-N-acetyl-glucosaminyltransferase *PmnagT* from *Pasteurella multocida* subsp. *multocida* str. HN06 (acc. no. PMCN06_0022) was codon-optimized and synthetically synthesized by GenScript Cooperation (Piscataway, USA). Cloning of the gene occurred by sequence and ligation-independent cloning into the plasmid pET-DUET (Merck KGaA, Darmstadt, Germany). All primer used for cloning are listed in table 3 below.

(88) *E. coli* BL21(DE3) 707 (table 2 below) harbouring plamid pET-PmnagT coding for a β -1,3-N-

acetyl glucosaminyltransferase was grown at 30° C. in mineral salts medium (Samain et al., J. Biotech. 1999, 72:33-47) supplemented with 2% (wt/vol) glucose and ampicillin 100 µg ml.^{sup.}-1. When the cultures reached an OD₆₆₀ nm of 0.1, gene expression was induced by addition of 0.3 mM IPTG. After four hours of incubation 1.5 mM lactose was added. After an additional incubation for 24 hours at 30° C. in shaking flasks cells were harvested. LNT-2 was detected by thin layer chromatography. Therefore, cells were mechanically disrupted in a defined volume using glass beads. Subsequently, samples were applied on TLC Silica Gel 60 F.sub.254 (Merck KGaA, Darmstadt, Germany). The mobile phase was composed of acetone:butanol:acetic acid:water (35:35:7:23).

(89) The result of the TLC analysis is shown in FIG. 2. The formation of a compound showing the same migration rate as the trisaccharide standard LNT-II could be observed when the gene *P_{mnagT}* was overexpressed. The LNT-II production strain 724 served as a control (19). Standards for lactose (1) and LNT-II (2) are depicted. LNT-II product formation in the samples is marked by asterisks.

Example 3

Generation of an *E. coli* Lacto-N-Triose II Production Strain Overexpressing a Homologous Sugar Efflux Transporter

(90) The export of oligosaccharides produced in *E. coli* was proven to be a limiting factor during the fermentation process. However, trisaccharides like 2'-fucosyllactose and LNT-2 are translocated into the culture supernatant to some extent, thus probably encoding a working sugar efflux transporter. In order to improve the efflux of lacto-N-triose II (LNT-II; GluNAc(β1-3)Gal(β1-4)Glc), the *E. coli* BL21(DE3) strain 1326 (table 2 below) was used for the screening of a library of sugar efflux transporters (SET). Putative SET proteins from *E. coli* were amplified from genomic DNA of *E. coli* BL21 (DE3) and integrated into vector pINT by sequence and ligation-independent cloning. Using the example of the gene *yjhB*, the primer 2567, 2568, 2526 and 2443 were used, generating the plasmid pINT-*yjhB*.

(91) The primer sequences used for cloning are listed in table 3 below. *E. coli* BL21(DE3) 1326 harbouring plasmids encoding for 20 different *E. coli* transporters were grown at 30° C. in mineral salts medium (Samain et al., J. Biotech. 1999, 72:33-47) supplemented with 2% (wt/vol) glucose, ampicillin 100 µg ml.^{sup.}-1 and zeocin 40 µg ml.^{sup.}-1. When the cultures reached an OD₆₆₀ nm of 0.1, gene expression of the genes was induced by addition of 200 ng/ml anhydrotetracycline. After four hours of incubation 2.5 mM lactose was added. After an additional incubation for 24 and 48 hours at 30°C in shaking flasks the LNT-II concentration in the supernatant was determined by LC-MS.

(92) Mass analysis was performed by characteristic fragment ion detection using an LC Triple-Quadrupole MS detection system. Precursor ions are selected and analyzed in quadrupole 1, fragmentation takes place in the collision cell using nitrogen as CID gas, selection of fragment ions is performed in quadrupole 3.

(93) Lacto-N-tetraose (LNT (Gal(β1-3)GlcNAc(β1-3)Gal(β1-4)Glc)), LNT-II and Maltotriose (internal standard for quantification) were analyzed in ESI positive ionization mode. LNT forms an ion of m/z 708.3 [M+H.^{sup.}+], LNT-II an ion of m/z 546.1 [M+H.^{sup.}+] and Maltotriose an ion of m/z 522.0 [M+NH._{sub.4}.^{sub.4}.^{sup.}+]. Adduct formation of this carbohydrate [m/z 504.0] takes place with an ammonium ion (NH₄.^{sup.}+), resulting in mass shift of +18. Thus for Maltotriose a precursor ion of m/z 522.0 was selected. The precursor ion was further fragmented in the collision cell into the characteristic fragment ions m/z 487.1, m/z 325.0 and m/z 163.2. The molecular ion of LNT (m/z 708.3) was fragmented into m/z 546.3, m/z 528.3, m/z 366.2 and m/z 204.0. LNT-II (m/z 546.1) was fragmented into m/z 204.2, 186.0, 138.0 and 126.0 (see method description).

(94) Chromatographic separation of LNT and LNT-II was performed on a Luna NH._{sub.2} HPLC column (Phenomenex, Aschaffenburg, Germany). This was necessary due to partial fragmentation of LNT during ionization resulting in LNT-II signals affecting quantification results of the

individual carbohydrates.

(95) Only for the strain expressing the gene *yjhB*, an increased amount of LNT-2 in the culture supernatant was observed (see table 1 below).

(96) Table 1: Calculated concentrations of LNT-II in the culture supernatant of an *E. coli* BL21 (DE3) strain overexpressing *yjhB* and the reference strain.

(97) TABLE-US-00001 Calc. conc. after 24 h Calc. conc. after 48 h Analyte Sample of incubation [μ M] of incubation [μ M] RT 1326 751 1265 0.616 1326 pINT-*yjhB* 413 1975 0.609

Example 4

(98) Batch Fermentations of *E. coli* BL21(DE3) 724 Screening Various β -1,4-Galactosyltransferases

(99) The genes for the β -1,4-galactosyltransferases *lex1* from *Aggregatibacter aphrophilus* NJ8700 (acc. no. YP_003008647), *PmgalT7* from *Pasteurella multocida* subsp. *multocida* str. HN06 (acc. No. PMCN06_0021), *MsgalT8* from *Myxococcus stipitatus* DSM14675 (acc. no. MYSTI_04346), *KdgalT10* from *Kingella denitrificans* ATCC 33394 (acc. no. HMPREF9098_2407), *gatD* from *Pasteurella multocida* M1404 (acc. no. GQ444331), *BfgalT2* from *Bacterioidis fragilis* NCTC9343 (acc. no. BF9343 0585), *IsgD* from *Haemophilus influenza* (acc. no. AAA24981) and *HpgalT* from *Helicobacter pylori* (acc. no. AB035971) were codon-optimized and synthetically synthesized by GenScript Cooperation (Piscataway, USA). Cloning of the genes occurred by sequence and ligation-independent cloning (Li and Elledge, Nat Methods. 2007 March; 4(3):251-6.). Therefore, the plasmid pINT, harbouring the *malE* gene under control of an anhydrotetracycline-inducible promoter, was used, enabling the generation of a N-terminal fusion of the β -1,4-galactosyltransferase genes with *malE*. Solely, the β -1,4-galactosyltransferase encoding gene *waaX* from *Pectobacterium atrosepticum* JG10-08 (acc. no. ECA0154) was cloned into plasmid pACYC-Duet (Merck KGaA, Darmstadt, Germany). All primer used for cloning are listed in table 3 below.

(100) *E. coli* BL21(DE3) 724 (table 2 below) harbouring plamid pCDF-galE and a plasmid coding for the gene fusion of *malE* with a β -1,4-galactosyltransferase was grown at 30° C. in mineral salts medium (Samain et al., J. Biotech. 1999, 72:33-47) supplemented with 2% (wt/vol) glucose, ampicillin 100 μ g ml.sup.-1 and zeocin 40 μ g ml.sup.-1. When the cultures reached an OD660 nm of 0.1, gene expression of the *galE* gene and the β -1,4-galactosyltransferase was induced by addition of 0.3 mM IPTG and 200 ng/ml anhydrotetracycline. *E. coli* BL21(DE3) 534 (table 2 below) harbouring plamids pET-*lgtA*, pCOLA-*glmUM-glmS*, pCDF-*galT-galE* and pACYC-*waaX* was grown at 30° C. in mineral salts medium supplemented with 2% (wt/vol) glucose, ampicillin 100 μ g ml.sup.-1, chloramphenicol 34 μ g ml.sup.-1, streptomycin 50 μ g ml.sup.-1 and kanamycin 30 μ g ml.sup.-1. When the cultures reached an OD660 nm of 0.1, gene expression was induced by addition of 0.3 mM IPTG. Four hours after induction of gene expression 2 mM lactose were added. After an additional incubation for 48 hours at 30° C. in shaking flasks, cells were harvested and mechanically disrupted. Lacto-N-neotetraose (LNnT (Gal(β 1-4)GlcNAc(β 1-3)Gal(β 1-4)Glc)) was detected by thin layer chromatography. Therefore, cells were mechanically disrupted using glass beads. Subsequently, samples were applied on TLC Silica Gel 60 F.sub.254 (Merck KGaA, Darmstadt, Germany). The mobile phase was composed of acetone:butanol:acetic acid:water (35:35:7:23).

(101) The results of the TLC analyses are shown in FIGS. 3-5. FIG. 3 shows the TLC analysis of culture extracts of lacto-N-tetraose (LNT) producing *E. coli* BL21(DE3) strains overexpressing the β -1,4-galactosyltransferase encoding genes *BfgalT2* (1), *PmgalT7* (3), *MsgalT8* (6), *gatD* (7), *lex1* (9), *IgtB* (11) or *IsgD* (13). Standards for lactose (15), LNT-II (16) and LNnT (17) are depicted. LNnT product formation in the samples is marked by asterisks.

(102) FIG. 4 shows the TLC analysis of culture extracts of lacto-N-tetraose (LNT) producing *E. coli* BL21(DE3) strains overexpressing the β -1,4-galactosyltransferase encoding genes *KdgalT10* (1), *cpsI14J* (7), *cpslaJ* (8, 9), *HpgalT* (12). Standards for lactose (3, 15), LNT-II (4, 16) and LNnT (5, 17) are depicted. LNnT product formation in the samples is marked by asterisks.

(103) FIG. 5 shows the TLC analysis of culture extracts of lacto-N-tetraose producing *E. coli* BL21(DE3) strains overexpressing the β -1,4-galactosyltransferase encoding gene *waaX* (5). Standards for lactose (1), LNT-II and LNT (2) are depicted. Again, LNT product formation in the samples is marked by asterisks.

(104) The formation of a compound showing the same migration rate as the tetrasaccharide standard LNT could be observed when the following genes were overexpressed: *lex1*, *PmgalT7*, *MsgalT8*, *BfgalT2*, *gatD*, *IsgD*, *KdgalT10*, *HpgalT*, *wax*.

(105) The β -1,4-galactosyltransferases *cpsIaJ* and *cpsI14J*, known from literature to produce LNT (Watanabe et al., J Biochem. 2002 February; 131(2):183-91; Kolkman et al., J Bacteriol. 1996 July; 178(13):3736-41), were also included in the activity screening and served as positive control. Using the described expression system, the formation of LNT could be observed by *CpsIaJ* and *CpsI14J* (FIG. 3). In total, 11 out of 30 tested genes were observed to produce LNT from LNT-II and UDP-galactose.

Example 5

(106) Batch Fermentations of *E. coli* BL21(DE3) 534 Screening Different β -1,3-Galactosyltransferases

(107) Using genomic DNA of *E. coli* K12 DH5a as template, *galE* was amplified using primer 1163 and 1162. The PCR product was purified, restricted with restriction endonucleases *NdeI* and *XhoI* and ligated into the second multiple cloning site of vector pCDFDuet (Merck KGaA, Darmstadt, Germany), which was cut with the same enzymes. *GalE* is expressed from the IPTG inducible T7 promoter. The *E. coli* K12 gene *galT* was amplified from genomic DNA and integrated into plasmid pCDF-*galE* by sequence and ligation-independent cloning using primer 991-994, producing the plasmid pCDF-*galT-galE*.

(108) Using the codon-optimized gene of *lgtA* as template, amplification occurred using primer 688 and 689. The PCR product was purified, restricted with restriction endonucleases *NdeI* and *AatII* and ligated into the multiple cloning site of vector pETDuet (Merck KGaA, Darmstadt, Germany), which was cut with the same enzymes, producing the plasmid pET-*lgtA*.

(109) Cloning of the codon-optimized gene construct of *glmUM* occurred by sequence and ligation-independent cloning into the plasmid pCOLA-Duet (Merck KGaA, Darmstadt, Germany) using primer 848-851. The codon-optimized form of *glmS* was amplified using primer 852 and 853. The PCR product was purified, restricted with restriction endonucleases *NdeI* and *AatII* and ligated into the second multiple cloning site of vector pCOLA-*glmUM*, which was cut with the same enzymes, producing the plasmid pCOLA-*glmUM-glmS*.

(110) The genes for the β -1,3-galactosyltransferases *wbdO* from *Salmonella enterica* subsp. *salamae* serovar Greenside (acc. no. AY730594) and *furA* from *Luticola nitroferum* 2002 (FuraDRAFT_0419) were also codon-optimized and synthetically synthesized by GenScript Cooperation (Piscataway, USA). Cloning of the genes occurred by sequence and ligation-independent cloning into the plasmid pACYC-Duet (Merck KGaA, Darmstadt, Germany). All primer used for cloning are listed in table 3 below.

(111) *E. coli* BL21(DE3) 534 harbouring plasmids pET-*lgtA*, pCOLA-*glmUM-glmS*, pCDF-*galT-galE* and a plasmid coding for a β -1,3-galactosyltransferase pACYC-*furA* or pACYC-*wbdO* was grown at 30° C. in mineral salts medium (Samain et al., J. Biotech. 1999, 72:33-47) supplemented with 2% (w/v) glucose, ampicillin 100 μ g ml.⁻¹, chloramphenicol 34 μ g ml.⁻¹, streptomycin 50 μ g ml.⁻¹ and kanamycin 30 μ g ml.⁻¹. When the cultures reached an OD₆₆₀ nm of 0.1, gene expression was induced by addition of 0.3 mM IPTG. After four hours of incubation 2 mM lactose was added. After an additional incubation for 48 hours at 30° C. in shaking flasks, cells were harvested. LNT was detected by thin layer chromatography. Therefore, cells were mechanically disrupted using glass beads. Subsequently, samples were applied on TLC Silica Gel 60 F.sub.254 (Merck KGaA, Darmstadt, Germany). The mobile phase was composed of acetone:butanol:acetic acid:water (35:35:7:23).

(112) The results of the TLC analyses are shown in FIG. 6, showing TLC analysis of culture extracts of lacto-N-tetraose producing *E. coli* BL21(DE3) strains overexpressing the β -1,3-galactosyltransferase encoding genes *wbdO* or *furA*. LNT product formation in the samples is marked. Out of 12 tested putative β -1,3-galactosyltransferases, the formation of a compound showing the same migration rate as the tetrasaccharide standard LNT could only be observed when genes *wbdO* and *furA* were overexpressed.

Example 6

(113) Development of an Improved Plasmid-Free *E. coli* Lacto-N-Tetraose Production Strain

(114) *Escherichia coli* BL21(DE3) strain 724 was used to construct a lacto-N-tetraose (LNT) producing strain. Metabolic engineering included the genomic integration of the transposon cassettes <P.sub.tet-*wbdO*-P.sub.T5-galE-FRT-cat-FRT> (SeqID5), flanked by the inverted terminal repeats specifically recognized by the mariner-like element HimarI transposase, which was inserted from pEcomar-*wbdO*-galE. The resulting strain 1353 was further metabolically engineered to exhibit an increased intracellular LNT-II pool resulting in the elevated production of LNT.

Therefore, the mayor facilitator superfamily transporter *yjhB* (acc. no. YP_003001824) was deleted from the genome of the *E. coli* strain, generating strain 1431 (table 2 below).

(115) Batch fermentation of the *E. coli* BL21(DE3) strains 1353 and 1431 was conducted for 48 hours at 30° C. in mineral salts medium (Samain et al., J. Biotech. 1999, 72:33-47) containing 2% (wt/vol) glucose as sole carbon and energy source. When the cultures reached an OD₆₆₀ nm of 0.5, 2.5 mM lactose was added. The presence of LNT-II and LNT in the culture supernatant was detected by high performance liquid chromatography (HPLC).

(116) Analysis by HPLC was performed using a refractive index detector (RID-10A) (Shimadzu, Duisburg, Germany) and a ReproSil Carbohydrate, 5 μ m (250 mm $\times$ 4.6 mm) (Dr. Maisch GmbH, Germany) connected to an HPLC system (Shimadzu, Duisburg, Germany). Elution was performed isocratically with acetonitril:H.sub.2O (68/32 (v/v)) as eluent at 35° C. and a flow rate of 1.4 ml/min. 40 μ l of the sample were applied to the column. Samples were filtered (0.22 μ m pore size) and cleared by solid phase extraction on an ion exchange matrix (Strata ABW, Phenomenex, Aschaffenburg, Germany).

(117) The results of the HPLC analyses are shown in FIG. 7, showing HPLC analyses of the culture supernatant of lacto-N-tetraose producing *E. coli* BL21 (DE3) strain. (A) Supernatant of *E. coli* BL21(DE3) 1353 (black graph) and 1431 (pink graph) grown in the presence of glucose and lactose after 24 h of incubation. (B) Supernatant of *E. coli* BL21(DE3) 1353 (blue graph) and 1431 (brown graph) grown in the presence of glucose and lactose after 48 h of incubation. As can be seen from the HPLC analyses, the deletion of *yjhB* in a LNT producing strain resulted in an elevated accumulation of LNT in the culture supernatant.

Example 7

(118) Generation of an *E. coli* Lacto-N-Tetraose Production Strain Overexpressing a Sugar Efflux Transporter

(119) Since an export of lacto-N-tetraose into the medium is only moderate for production strains, a screening of a sugar efflux transporter library was conducted. In accordance to example 3 putative SET proteins were either amplified from *E. coli* genomic DNA or were codon-optimized and synthetically synthesized by GenScript Cooperation (Piscataway, USA). Following amplification genes were integrated into vector pINT by sequence and ligation-independent cloning. The primer design for the cloning of *E. coli* genes was in accordance to example 3. Synthetic genes were synthesized with standardized nucleotide overhangs and likewise integrated into the expression vector using the primer 2527, 2444, 2526 and 2443. The primer sequences used for cloning are listed in table 3 below.

(120) *E. coli* BL21(DE3) 1353 (table 2 below) harbouring plasmids encoding for 66 different transporters were grown at 30° C. in mineral salts medium (Samain et al., J. Biotech. 1999, 72:33-47) supplemented with 3% (w/v) glucose, 5 g l.sup.-1 NH.sub.4Cl.sub.2, ampicillin 100 μ g ml.sup.-1.

-1 and kanamycin 15 µg ml.sup.-1. Precultivation appeared in 96-well plates harbouring a total volume of 200 µl. After 24 h of incubation at 30° C. by continuous shaking, 50 µl per well was transferred into 96-well deep well plates harbouring a total volume of 400 µl mineral salts medium additionally supplemented with 200 ng ml.sup.-1 anhydrotetracycline and 10 mM lactose. After a sustained incubation for 24 to 48 hours the LNT concentrations in the supernatant were determined by LC-MS. Mass analysis was performed as described in example 3.

(121) FIG. 8 shows the relative concentration of lacto-N-tetraose in the supernatant of *E. coli* BL21 (DE3) strains overexpressing sugar efflux transporters compared to the control strain 1353. The LNT titer of strain 1353 was set to 100%. As shown in FIG. 8, the overexpression of 11 out of 66 genes resulted in a doubled LNT production. Among these, also a protein encoded in the genome of *E. coli* BL21 (DE3) proved to enhance the LNT export (TP37, yebQ, acc. no. NC_012971). YebQ is a predicted MFS transporter, putatively involved in multi drug efflux, which might represent a responsible transporter protein that realizes the observed basal efflux of LNT during fermentation of strain 1353.

(122) Furthermore, the exporters encoded by the genes spoVB of *Bacillus amyloliquefaciens* (TP1, acc. no. AFJ60154), yabM of *Erwinia pyrrilfoliae* (TP2, acc. no. CAY73138), bcr of *E. coli* MG1655 (TP18, acc. no. AAC75243), ydeA of *E. coli* MG1655 (TP20, acc. no. AAC74601), proP2 of *Haemophilus parainfluenzae* (TP54, acc. no. EGC72107), setA of *Pectobacterium carotovorum* (TP55, acc. no. ZP_03829909), fucP of *E. coli* MG1655 (TP59, acc. no. AIZ90162), mdeA of *Staphylococcus aureus* Bmb9393 (TP61, acc. no. SABB_01261), lmrA of *Lactococcus lactis* (TP62, acc. no. L116532), setA of *Pseudomonas* sp. MT-1 (TP72, acc. no. BAP78849) and setA of *Beauveria bassiana* D1-5 (TP73, acc. no. KGQ13398) resulted in an increased LNT production when overexpressed in the *E. coli* production strain 1353.

Example 8

(123) Generation of an *E. coli* Lacto-N-Triose II Production Strain by Overexpression of Heterologous Sugar Efflux Transporters

(124) The LNT exporter screening described in example 6 interestingly disclosed two proteins—TP11 from *Mannheimia succiniciproducens* MBEL55E (proP, acc. no. AAU37785) and TP70 from *Cedecea neteri* M006 (setA, acc. no. WP_039290253)—whose overexpression resulted in a significantly increased production of LNT-II and consequently in a decreased LNT production (data not shown). This observation was confirmed in an experimental setup as described in example 3. The overexpression of the sugar efflux transporter YjhB served as a positive control. The overexpression of TP11 as well as TP70 resulted in an approximately 4-fold increase in LNT-II production which was even slightly more than for YjhB: FIG. 9 shows a diagram displaying the concentrations of lacto-N-triose II in the supernatant of *E. coli* BL21 (DE3) strains overexpressing the sugar efflux transporters TP11 (2), YjhB (3) or TP70 (4). Strain 1326 harbouring an empty control plasmid served as a control (1). Thus, 3 sugar efflux transporters were identified which target LNT-II for export and whose overexpression might be useful to engineer a LNT-II production strain.

(125) TABLE-US-00002 TABLE 2 Strains and plasmids Strain Genotype Ref. *E. coli* BL21(DE3) F-ompT hsdSB(rB-, mB-) gal dcm (DE3) Merck KGaA, Darmstadt, Germany *E. coli* BL21(DE3) 534 *E. coli* BL21(DE3) ΔlacZ Δara ΔwcaJ This study ΔfucIK ΔnagAB harbouring genomic integrations of: galETKM, lacY *E. coli* BL21(DE3) 724 *E. coli* BL21(DE3) ΔlacZ Δara ΔwcaJ This study ΔfucIK ΔnagAB harbouring genomic integrations of: galETKM, lacY, lgtA-galT- kanR, glmUM-glmS-dhfr *E. coli* BL21(DE3) 1326 *E. coli* BL21(DE3) ΔlacZ Δara ΔwcaJ This study ΔfucIK ΔnagAB harbouring genomic integrations of: galETKM, lacY, lgtA-galT- kanR, glmUM-glmS-dhfr, lacy(6HIS)-aadA *E. coli* BL21(DE3) 707 *E. coli* BL21(DE3) ΔlacZ Δara ΔwcaJ This study ΔfucIK ΔnagAB harbouring genomic integrations of: galETKM, lacY, glmUM- glmS-dhfr *E. coli* BL21(DE3) 1353 *E. coli* BL21(DE3) ΔlacZ Δara ΔwcaJ This study ΔfucIK ΔnagAB harbouring genomic integrations of: galETKM, lacY, lgtA-galT- kanR, glmUM-glmS-dhfr, wbdO-

galE-cat *E. coli* BL21(DE3) 1431 *E. coli* BL21(DE3) ΔlacZ Δara ΔwcaJ This study ΔfucLK
ΔnagAB harbouring genomic integrations of: galETKM, lacY, lgtA-galT- kanR, glmUM-glmS-
dhfr, wbdO-galE-cat, ΔyjhB-aacC1 pCDF-galE galE of *E. coli* K12 integrated into vector EP 14
162 869.3 pCDFDuet pET-lgtA (SeqID7) lgtA of *Neisseria meningitidis* integrated This study into
vector pETDuet pCDF-galT-galE (SeqID8) galT and galE of *Escherichia coli* K12 This study
integrated into vector pCDFDuet pCOLA-glmUM-glmS glmU, glmM and glmS of *Escherichia*
coli This study (SeqID9) K12 integrated into vector pCOLADuet pINT-male-lex1 Gene fusion of
maleE with lex-1 of EP 14 162 869.3 *Aggregatibacter aphrophilus* NJ8700 integrated into vector
pINT pINT-male-PmgalT7 Gene fusion of PmgalT7 of *Pasteurella* This study (SeqID10)
multocida subsp. *multocida* str. HN06 integrated into vector pINT pINT-male-MsgalT8 Gene
fusion of MsgalT8 of *Myxococcus* This study (SeqID11) *stipitatus* DSM14675 integrated into
vector pINT pINT-male-KdgalT10 Gene fusion of KdgalT10 of *Kingella* This study (SeqID12)
denitrificans ATCC 33394 integrated into vector pINT pINT-male-gatD (SeqID13) Gene fusion of
gatD of *Pasteurella* This study *multocida* M1404 integrated into vector pINT pINT-male-BFgalT2
Gene fusion of BFgalT2 of *Bacterioidis* This study (SeqID14) *fragilis* NCTC9343 integrated into
vector pINT pINT-male-IsgD (SeqID15) Gene fusion of IsgD of *Haemophilus* This study *influenza*
integrated into vector pINT pINT-male-HPgalT Gene fusion of HpgalT of *Helicobacter* This study
(SeqID16) *pylori* integrated into vector pINT pACYC-waaX (SeqID17) waaX of *Pectobacterium*
atrosepticum This study JG10-08 integrated into vector pACYCDuet pACYC-wbdO (SeqID18)
wbdO of *Salmonella enterica* subsp. This study *salamae* serovar Greenside integrated into vector
pACYCDuet pACYC-furA (SeqID19) furA of *Lutiella nitroferum* 2002 integrated This study into
vector pACYCDuet pET-PmnagT (SeqID20) PmnagT of *Pasteurella multocida* subsp. This study
multocida str. HN06 integrated into vector pETDuet PINT-yjhB (SeqID21) yjhB of *E. coli* BL21
DE3 integrated into This study vector pINT pINT-yebQ (SeqID22) yebQ of *E. coli* BL21 DE3
integrated into This study vector pINT pINT-proP (SeqID23) proP of *Mannheimia*
succiniciproducens This study MBEL55E integrated into vector pINT pINT-Cn-setA (SeqID24)
setA of *Cedecea neteri* M006 integrated This study into vector pINT pINT-spoVB (SeqID25)
spoVB of *Bacillus amyloliquefaciens* This study integrated into vector pINT pINT-yabM
(SeqID26) yabM of *Erwinia pyrifoliae* integrated into This study vector pINT pINT-ydeA
(SeqID27) ydeA of *E. coli* MG1655 integrated into This study vector pINT pINT-proP2 (SeqID28)
proP2 of *Haemophilus parainfluenzae* This study integrated into vector pINT pINT-Pc-setA
(SeqID29) setA of *Pectobacterium carotovorum* This study integrated into vector pINT pINT-fucP
(SeqID30) fucP of *Escherichia coli* BL21 (DE3) This study integrated into vector pINT pINT-
mdeA (SeqID31) mdeA of *Staphylococcus aureus* Bmb9393 This study integrated into vector pINT
pINT-lmrA (SeqID32) lmrA of *Lactococcus lactis* integrated into This study vector pINT pINT-Ps-
setA (SeqID33) setA of *Pseudomonas* sp. MT-1 integrated This study into vector pINT pINT-Bb-
setA (SeqID34) setA of *Beauveria bassiana* D1-5 This study integrated into vector pINT
(126) TABLE-US-00003 TABLE 3 Oligonucleotides used for PCR Primer Sequence 5'-3'
605 KI gal fwd TTA CT CAG CAATAAACTGATATTC CGTCAGGCTGG (SeqID35) 606
KI gal rev TTGTAATCTCGCGCTCTTCACATCAGACTTTC CATATAGAGCGTAATTTC
CGTTAACGTCGGTAGTGCTGACCTTGCCGGAGG (SeqID36) 1119 ME-for
CTGTCTCTTATCACATCTCCTGAAATGGCCAGATGTAATTCCTAATTTTT GTTG
(SeqID37) 1120 ME rev
CTGTCTCTTATCACATCTCACATTACATCTGAGCGATTGTTAGG (SeqID38) 1163
galE_NdeI-for GATCACATATGAGAGTTCTGGTTACCGGTG (SeqID39) 1164 galE_XhoI-
rev GATCACTCGAGTCATTAATCGGGATATCCCTGTGGATGGC (SeqID40) 5176 lex1
pINT-f GTCGATGAAGCCCTGAAAGACGCGCAGACTATGCACTTCATTGAAAAC
AAAAACTTCGTC (SeqID41) 5177 lex1 pINT-r
GATGGCCTTTTTGCGTGTCGACGCGGCCGCTAGATAAACAGGATGAT
ATTTTTGCCTIG (SeqID42) 5178 pINT lex1-f

CAAGGCAAAATATCATCTGTTTATCTAGCGCGCGTCGACACGC
AAAAAGGCCATC (SeqID43) 5179 pINT lexl-r
GACGAAGTTTTTTGTTTTCAATGAAGTGCATAGTCTGCGCGTCTTTCAGG
GCTTCATCGAC (SeqID44) 5192 waaX pINT for
GTCGATGAAGCCCTGAAAGACGCGCAGACTATGATTGATAACCTGATTA
AGCGTACCCCG (SeqID45) 5193 waaX pINT rev
ATGGCCTTTTTTGCGTGTCGACGCGGCCGCTTAATTCGAGCGGGTAAAG
ATCTTCATCAGG (SeqID46) 5194 pINT waaX for
CTGATGAAGATCTTTACCCGCTCGAATTAAGCGGCCGCGTCGACACGC
AAAAAGGCCATC (SeqID47) 5195 pINT waaX rev
CGGGGTACGCTTAATCAGGTTATCAATCATAGTCTGCGCGTCTTTCAGG
GCTTCATCGAC (SeqID48) 5164 PmgalT7 pINT
GTCGATGAAGCCCTGAAAGACGCGCAGACTATGAGCGGTGAACACTAT for
GTCATTAGCCTG (SeqID49) 5165 PmgalT7 pINT
GATGGCCTTTTTTGCGTGTCGACGCGGCCGCTCATTAAATTCGATGATC rev
ATCTTGTCGTT (SeqID50) 5166 pINT PmgalT7
AACGACAAGATGATCATCGAATTTAAATGAGCGGCCGCGTCGACACGC for
AAAAAGGCCATC (SeqID51) 5167 pINT PmgalT7
CAGGCTAATGACATAGTGTTACCGCTCATAGTCTGCGCGTCTTTCAGG rev
GCTTCATCGAC (SeqID52) 5168 MsgalT8 pINT
GTCGATGAAGCCCTGAAAGACGCGCAGACTATGGATGAAATCAAACCTG for
TCGGTGGTTATG (SeqID53) 5169 MsgalT8 pINT
GATGGCCTTTTTTGCGTGTCGACGCGGCCGCTCATTGGCGACGCCAATC rev
GAACGCAACGCG (SeqID54) 5170 pINT MsgalT8
CGCGTTGCGTTCGATTGGCGTCGCCAATGAGCGGCCGCGTCGACACG for
CAAAAAGGCCATC (SeqID55) 5171 pINT MsgalT8
CATAACCACCGACAGTTTGATTTCATCCATAGTCTGCGCGTCTTTCAGG rev
GCTTCATCGAC (SeqID56) 5561 KdgalT10
GTCGATGAAGCCCTGAAAGACGCGCAGACTATGGAAAACCTATGTCGTC pINT for
TCTATCCGCACC (SeqID57) 5562 KdgalT10
GATGGCCTTTTTTGCGTGTCGACGCGGCCGCTCATTGAAACGGAACAAT pINT-rev
CTTTTTGTCATC (SeqID58) 5563 pINT-
GATGACAAAAAGATTGTTCCGTTCAAATGAGCGGCCGCGTCGACACGC KdgalT10 for
AAAAAGGCCATC (SeqID59) 5564 pINT-
GGTGCGGATAGAGACGACATAGTTTTCCATAGTCTGCGCGTCTTTCAG KdgalT10 rev
GGCTTCATCGAC (SeqID60) 5172 gatD pINT for
GTCGATGAAGCCCTGAAAGACGCGCAGACTATGTCCTCAGCTTTCCATT
ACGTCATTAGC (SeqID61) 5173 gatD pINT rev
GATGGCCTTTTTTGCGTGTCGACGCGGCCGCTCATTCAAATTCGATAATC
ATGGTGATTTT (SeqID62) 5174 pINT gatD for
AAAATCACCATGATTATCGAATTTGAATGAGCGGCCGCGTCGACACGCA
AAAAGGCCATC (SeqID63) 5175 pINT gatD rev
GCTAATGACGTAATGGAAAGCTGAGGACATAGTCTGCGCGTCTTTCAG
GGCTTCATCGAC (SeqID64) 5160 BfgalT2 pINT
GTCGATGAAGCCCTGAAAGACGCGCAGACTATGAACGTGAATAAGCCG for
ACCACCGAAAAG (SeqID65) 5161 BfgalT2 pINT
GATGGCCTTTTTTGCGTGTCGACGCGGCCGCTCAGTATTCTTCAATTTTG rev
TCCAGTTGATA (SeqID66) 5162 pINT BfgalT2
TATCAACTGGACAAAATTGAAGAATACTGAGCGGCCGCGTCGACACGC for
AAAAAGGCCATC (SeqID67) 5163 pINT BfgalT2

CTTTTCGGTCTGATTCTACGTTTCATAGTCTGCGCGTCTTTCAGG rev
GCTTCATCGAC (SeqID68) 5746
GTGATCAACGCCGCCAGCGGTCGTCAGACTGTCGATGAAGCCCTGAAA
GACGCGCAGACT (SeqID69) 5747
GCGGCCGCGTTCGACACGCAAAAAGGCCATCCATCCGTCAGGATGGCC
TTCTGCTTAATTT (SeqID70) 5748
AAATTAAGCAGAAGGCCATCCTGACGGATGGATGGCCTTTTTGCGTGT
CGACGCGGCCGC (SeqID71) 5749
AGTCTGCGCGTCTTTCAGGGCTTCATCGACAGTCTGACGACCGCTGGC
GGCGTTGATCAC (SeqID72) 1886 SLIC wbdO
GTTTAACTTTAATAAGGAGATATAACCATGCTGACGGAAGTGC GCCCGGT pACYC for
CTCTACGACGAAACCGC (SeqID73) 1887 SLIC wbdO
CGACCTGCAGGCGCGCCGAGCTCGAATTCATTTGATGTATTTGCAATA pACYC rev
GAACACAGAAAAGACCGT (SeqID74) 1888 SLIC pACYC
GTGTTCTATTGCAAATACATCAAATGAATTCGAGCTCGGCGCGCCTGCA wbdO rev
GGTCGACAAGCTTGCGG (SeqID75) 1889 SLIC pACYC
GAGACCGGGCGCACTTCCGTCAGCATGGTATATCTCCTTATTAAAGTTA WbdO For
AACAAAATTATTTCTACAGG (SeqID76) 1890 SLIC pACYC
GTATGGTGACCCTGTGGCGCAAATGAGAATTCGAGCTCGGCGCGCCTG furA rev
CAGGTGACAAGCT (SeqID77) 1891 SLIC pACYC
GCGCTGCCCTGTTTGATTTTATCCATGGTATATCTCCTTATTAAAGTTAA furA for
ACAAAATTATTTCT (SeqID78) 1892 SLIC furA
CCTGCAGGCGCGCCGAGCTCGAATTCTCATTTGCGCCACAGGGTCACC pACYC rev
ATACGTGCCGGCAGG (SeqID79) 1893 SLIC furA
GITTAACTTTAATAAGGAGATATAACCATGGATAAAATCAAACAGGGCAG pACYC for
CGCCTCTCTGGTTGTCG (SeqID80) 3055 SLIC PmnagT
CAGACTCGAGGGTACCGACGTCCTAATAAGTAGATGAATATTTATCAGG pET rev
ACGAAGAT (SeqID81) 3056 SLIC pET
AACTAAAGGTTTATTTTCCATATGTATATCTCCTTCTTATACTTAACTAAT PmnagT for
ATAC (SeqID82) 3057 SLIC pET
TAAATATTCATCTACTTATTAGGACGTCGGTACCCTCGAGTCTGGTAAA PmnagT rev
GAAACCGCTGCTGCG (SeqID83) 3058 SLIC PmnagT
GTATAAGAAGGAGATATAACATATGGAAAATAAACCTTTAGTTTCAGTTTT pET for
GATTTGTGC (SeqID84) 2567_SLIC_yjhB-for
TAACTTTAAGAAGGAGATATAACAAGAGCTCGAGTCGAAGGAGATAGAAC
CATGGCAACAGCATGGTATAACAAG (SeqID85) 2568_SLIC_yjhB-
GCGTGTCGACGCGTTTAGAGGCCCAAGGGGTTATGCTAGTATCGATT rev
TATCATTTAGCCACGGATAGTTTATAAATTTTAC (SeqID86) 2526_SLIC_pINT_
GGTTCTATCTCCTTCGACTCGAGCTCTTGTATATCTCCTTCTTAAAGTTA TP-rev
AACAAAATTATTTCTAGATTTTTGTCGAAC (SeqID87) 2443_SLIC_pINT_
TAAATCGATACTAGCATAACCCCTTGGGGCCTCTAAACGCGTCGACAC TP-forw
GCAAAAAGGCCATCC (SeqID88) 2527_SLIC_TP_pINT-
GTTCGACAAAAATCTAGAAATAATTTTGTTTAACTTTAAGAAGGAGATAT forw
ACAAGAGCTCGAGTCGAAGGAGATAGAACC (SeqID89) 2444_SLIC_TP_pINT-
GGATGGCCTTTTTGCGTGTCGACGCGTTTAGAGGCCCAAGGGGTTAT rev
GCTAGTATCGATTTA (SeqID90) 688 IgtA AatII rev
ATATGACGTCTCATTAGCGGTTTTTCAGGAGACG (SeqID91) 689 IgtA NdeI for
ATATCATATGCCGTCCGAAGCATTCCGTCGTCACC (SeqID92) 991 galT-pCDF for
TAACTTTAATAAGGAGATATAACCATGACGCAATTTAATCCCGTTGATCAT
CCACATCGCCGC (SeqID93) 992 pCDF-galT for

ATTTTCGCGAATCCGGAGTGTTAAAGCTTGCGGCCGCATAATGCTTAAG
 TCGAACAGAAAGTAATCG (SeqID94) 993 galT-pCDF rev
 AAGCATTATGCGGCCGCAAGCTTTTACACTCCGGATTCGCGAAAATGG
 ATATCGCTGACTGCGCGCAAACGC (SeqID95) 994 pCDF-galT rev
 TCAACGGGATTAAATTGCGTCATGGTATATCTCCTTATTAAAGTTAAACA
 AAATTATTTCTACAGGGG (SeqID96) 848 glmM pCOLA
 ATGGTGATGGCTGCTGCCCATTTAAACCGCTTTGACTGCGTCGGCAATA SLIC rev
 CGGTGCGC (SeqID97) 849 glmU pCOLA
 GTTTAACTTTAATAAGGAGATATACCATGCTGAACAACGCGATGTCTGTT SLIC for
 GTTATCCTGG (SeqID98) 850 pCOLA glmM
 CGCAGTCAAAGCGGTTTAAATGGGCAGCAGCCATCACCATCATCACCA SLIC rev
 CAGCC (SeqID99) 851 pCOLA glmU
 TCGCGTTGTTTCAGCATGGTATATCTCCTTATTAAAGTTAAACAAAATTAT SLIC for
 TTCTACAGG (SeqID100) 852 glmSco pCOLA
 ATATATCATATGTGCGGTATCGTTGGTGCTATCGC (SeqID101) for NdeI 853 glmSco
 pCOLA ATATATGACGTCTTATTCCACGGTCACGGATTTCGC (SeqID102) rev AatII

Claims

1. A method for production of an oligosaccharide by a genetically modified microbial host cell, wherein the oligosaccharide is a lacto-N-tetraose comprising a lacto-N-triose II (LNT-II; GlcNAc(b1-3)Gal(b1-4)Gluc) as a core trisaccharide, the method comprising: providing a genetically modified microbial host cell that comprises: at least one recombinant glycosyltransferase selected from the group consisting of a β -1,3-N-acetylglucosaminyltransferase and a β -1,3-galactosyltransferase; increased expression or activity of at least one homologous or heterologous sugar export protein capable of exporting the oligosaccharide; and decreased expression or inactivation of at least one export protein that exports LNT-II from the host cell; cultivating the host cell in a medium under conditions permissive for the production of the oligosaccharide, whereby the oligosaccharide is exported into the medium at an increased level compared to an unmodified host cell, and obtaining the oligosaccharide from the medium.
2. The method of claim 1, wherein the host cell further comprises: overexpression of at least one homologous or heterologous nucleic acid sequence coding for the sugar export protein capable of exporting the oligosaccharide into the culture medium; and/or deletion, disruption, diminishment or inactivation of at least one endogenous nucleic acid sequence coding for an export protein that exports precursors of the oligosaccharide outside the host cell; and/or overexpression of at least one homologous or heterologous sequence coding for a protein mediating import of a precursor of the oligosaccharide into said host cell, wherein the precursor is larger than a disaccharide.
3. The method of claim 2, wherein said sugar export protein capable of exporting the oligosaccharide belongs to the class of secondary active transporters.
4. The method of claim 2, wherein the nucleic acid sequence coding for the homologous or heterologous sugar export protein capable of exporting the oligosaccharide is of bacterial, archeal, plant, yeast or animal origin.
5. The method of claim 2, wherein said at least one nucleic acid sequence coding for the sugar export protein that exports precursors of the oligosaccharide outside the host cell is a gene selected from the group consisting of yebQ from *Escherichia coli*, yjhB from *Escherichia coli*, proP from *Mannheimia succiniciproducens*, and setA from *Cedecea neteri*.
6. The method of claim 1, wherein said β -1,3-N-acetylglucosaminyltransferase belongs to the class of lgtA of *Neisseria meningitides* or PmnagT of *Pasteurella multocida*.
7. The method of claim 1, wherein the β -1,3-galactosyltransferase is encoded by a WbdO gene.
8. The method of claim 1, wherein-ii the genetically modified microbial host cell further comprises

an endogenous β -galactosidase gene and glucosamine-6-phosphate deaminase gene that are inactivated or deleted, and wherein said genetically modified microbial host cell further comprises a nucleic acid sequence coding for a functional lactose permease protein.

9. The method of claim 1, wherein the genetically modified microbial host cell further comprises an increased UDP-N-acetylglucosamine and a UDP-galactose, or GDP-fucose, or CMP-N-acetylneuraminic acid production, as compared to a genetically unmodified host cell, wherein optionally said increased UDP-N-acetylglucosamine and UDP-galactose production capability is by an overexpression of one or more genes encoding a protein selected from the group consisting of L-glutamine:D-fructose-6-phosphate aminotransferase, N-acetyl glucosamine-1-phosphate uridyltransferase/glucosamine-1-phosphate acetyl transferase, phosphoglucosamine mutase, UDP-galactose-4-epimerase, phosphoglucomutase, and glucose-1-phosphate uridylyltransferase.

10. The method of claim 8, wherein said genetically modified microbial host cell is cultivated in the presence of glucose, sucrose, glycerol or a combination thereof.

11. The method of claim 10, wherein the microbial host cell is cultured in the absence of N-acetylglucosamine and galactose.

12. A genetically modified microbial host cell for the production of lacto-N-tetraose comprising a lacto-N-triose II (LNT-II; GlcNAc(β 1-3)Gal(β 1-4)Gluc) as a core trisaccharide, wherein the host cell comprises at least one recombinant glycosyltransferase, selected from the group consisting of a β -1,3-N-acetylglucosaminyltransferase and a β -1,3-galactosyltransferase; at least one homologous or heterologous nucleic acid sequence coding for a sugar export protein capable exporting of the lacto-N-tetraose into a culture medium the host cell is cultivated in wherein said sugar export protein capable of exporting the lacto-N-tetraose belongs to the class of secondary active transporters, and wherein said homologous or heterologous nucleic acid sequence coding for the sugar export protein capable of exporting the lacto-N-tetraose is overexpressed or under control of a promoter sequence enabling the overexpression of said nucleic acid sequence; and inactivated expression or activity of at least one endogenous sugar export protein that exports LNT-II from the host cell.

13. The microbial host cell of claim 12, further comprising an overexpression of at least one homologous or heterologous nucleic acid sequence coding for a protein mediating the import of a precursor of the lacto-N-tetraose into said host cell through genetic modification of the microbial host cell, wherein the precursor is larger than a disaccharide.

14. The microbial host cell of claim 12, wherein said nucleic acid sequence coding for the sugar export protein capable of exporting the lacto-N-tetraose is of bacterial, archeal, plant, yeast or animal origin.

15. The microbial host cell of claim 12, wherein said at least one nucleic acid sequence coding for the export protein that exports LNT-II from the host cell is a gene selected from the group consisting of yebQ from *Escherichia coli*, yjhB from *Escherichia coli*, proP from *Mannheimia succiniciproducens*, and setA from *Cedecea neteri*.

16. The microbial host cell of claim 14, wherein the sugar export protein capable of exporting the lacto-N-tetraose is selected from the group consisting of YebQ from *Escherichia coli* BL21(DE3), SpoVB of *Bacillus amyloliquefaciens*, YabM of *Erwinia pyrillifolia*, Bcr of *E. coli* MG1655, YdeA of *E. coli* MG1655, ProP2 of *Haemophilus parainfluenzae*, SetA of *Pectobacterium carotovorum*, FucP of *E. coli* MG1655, MdeA of *Staphylococcus aureus* Bmb9393, ImrA of *Lactococcus lactis*, SetA of *Pseudomonas* sp. MT-1 and SetA of *Beauveria bassiana* D1-5.

17. The method of claim 1, wherein the sugar export protein capable of exporting the lacto-N-tetraose is selected from the group consisting of YebQ from *Escherichia coli* BL21(DE3), SpoVB of *Bacillus amyloliquefaciens*, YabM of *Erwinia pyrillifolia*, Bcr of *E. coli* MG1655, YdeA of *E. coli* MG1655, ProP2 of *Haemophilus parainfluenzae*, SetA of *Pectobacterium carotovorum*, FucP of *E. coli* MG1655, MdeA of *Staphylococcus aureus* Bmb9393, ImrA of *Lactococcus lactis*, SetA of *Pseudomonas* sp. MT-1 and SetA of *Beauveria bassiana* D1-5.

18. The method of claim 1, wherein the microbial host cell is an *Escherichia coli* cell, a *Corynebacterium glutamicum* cell or a *Saccharomyces* sp. cell.

19. The microbial host cell of claim 12, wherein the microbial host cell is an *Escherichia coli* cell, a *Corynebacterium glutamicum* cell or a *Saccharomyces* sp. cell.
